

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2022

TEST DATE: 10-04-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A (One Options Correct Type)

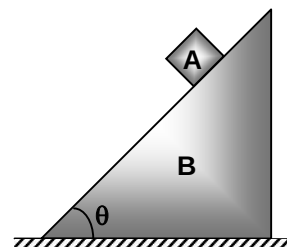
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The true value of angle of dip at place is 60° , the apparent dip in a plane inclined at an angle of 30° with magnetic meridian at this place is

- (A) $\tan^{-1}\left(\frac{1}{2}\right)$
 (B) $\tan^{-1}(2)$
 (C) $\tan^{-1}\left(\frac{2}{3}\right)$
 (D) none of these

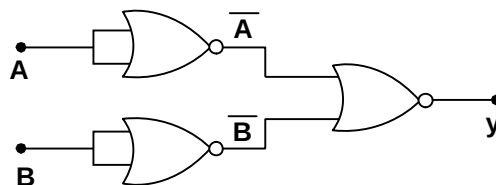
2. In the figure shown, A & B are free to move. All the surfaces are smooth. then : ($0 < \theta < 90^\circ$)

- (A) the acceleration of A will be more than $g \sin \theta$
 (B) the acceleration of A will be less than $g \sin \theta$
 (C) normal force on A due to B will be more than $mg \cos \theta$
 (D) normal force on A due to B will be equal to $mg \cos \theta$



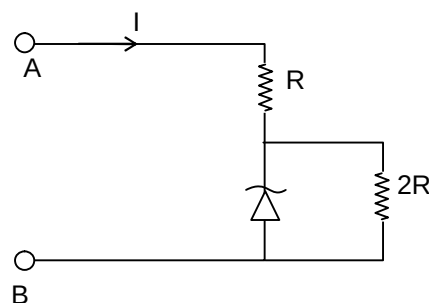
3. Identify the logic operation performed by the circuit given below

- (A) NOT
 (B) AND
 (C) OR
 (D) NAND



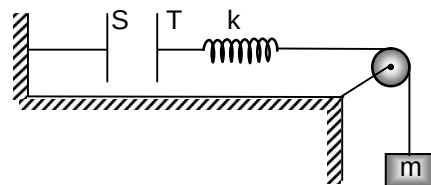
4. If the voltage between the terminals A and B is 17 V and zener breakdown voltage is 9V, then the potential voltage across R is

- (A) 6V
 (B) 8V
 (C) 9V
 (D) 17V



5. The plates S and T of an uncharged parallel plate capacitor are connected across battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on one of the capacitor plates is: [Area of plates = A]

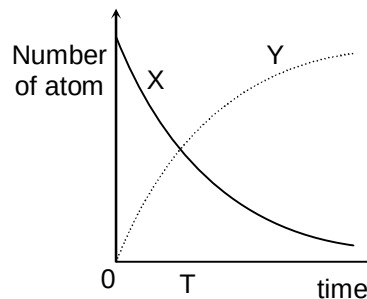
- (A) $\sqrt{2mgA\epsilon_0}$



- (B) $\sqrt{\frac{4mgA\varepsilon_0}{k}}$
 (C) $\sqrt{mgA\varepsilon_0}$
 (D) $\sqrt{\frac{2mgA\varepsilon_0}{k}}$

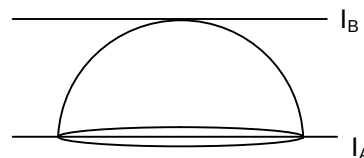
6. The graph represents the decay of a newly prepared sample of radioactive nuclide X to a stable nuclide Y. The half-life of X is τ . The growth curve for Y intersects the decay curve for X after time T. The time T is

- (A) $\frac{\tau}{2}$
 (B) $\ln\left(\frac{\tau}{2}\right)$
 (C) τ
 (D) 2τ



7. Moment of inertia of a uniform solid hemi-sphere about given axis is I_A and I_B then:

- (A) $I_A > I_B$
 (B) $I_A = I_B$
 (C) $I_A < I_B$
 (D) Can't be predicted



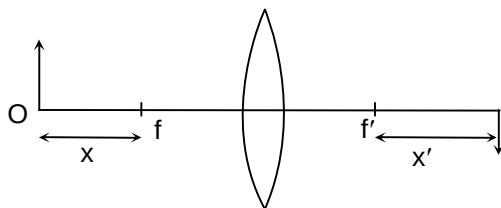
8. A hydrogen atom at rest emits a photon during its transition from $n = 2$ to $n = 1$. Choose the incorrect statement.

- (A) De-Broglie wavelength of hydrogen atom during recoil will be same as wavelength of emitted photon.
 (B) Energy of emitted photon will be less than 10.2 eV
 (C) Kinetic energy of recoil H-atom will be less than 10.2 eV
 (D) Kinetic energy of recoil H-atom will be equal to energy of emitted photon.

9. A monoatomic gas undergoes a process $T \cdot P^{(-1/3)} = \text{constant}$. If heat is given to the gas during the process, then:

- (A) Its temperature remains constant
 (B) Its temperature will increase
 (C) Its temperature will decrease
 (D) Can't be predicted

10. An object is placed at a point, which is at a distance x from the focus of a convex lens and its image is formed at I, which is at a distance x' from the focus f' (as shown in the figure). The distance x and x' satisfy the relation

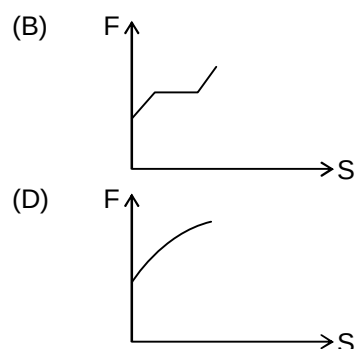
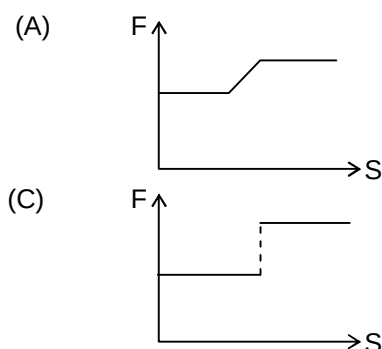
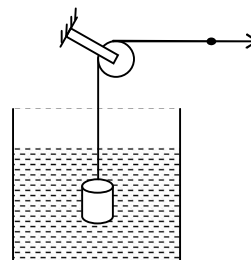


- (A) $\frac{x + x'}{2} = f$
 (B) $f = xx'$
 (C) $x + x' \leq 2f$
 (D) $x + x' \geq 2f$

11. An air bubble of diameter 2 mm rises steadily through a solution of density 1750 kg/m^3 at the rate of 0.35 cm/s . The coefficient of viscosity of the solution is nearly (Assume mass of the bubble to be negligible)

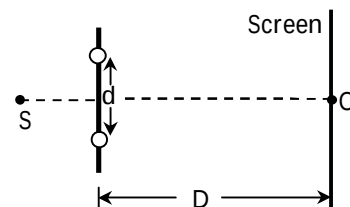
(A) 9 poise
(B) 6 poise
(C) 11 poise
(D) 4 poise

12. A metallic cylinder of radius r and height h is totally submerged in a liquid contained in a beaker as shown in the figure. If a force F is applied on the string to pull the cylinder with a constant velocity v , the graph showing the variation of force F against displacement of cylinder is best represented by

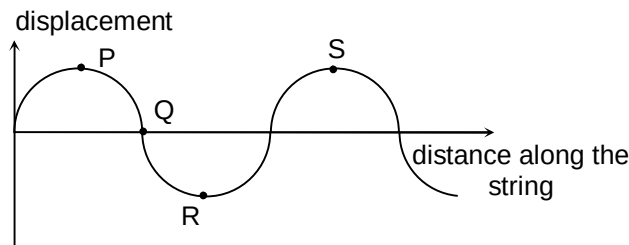


13. In a YDSE experiment the primary source S (which is far away from the slits) is moved towards slits symmetrically and slowly. As a result

(A) Fringe width increases
(B) Fringe width decreases
(C) Positions of maxims remains the same.
(D) Intensity of fringes decreases.



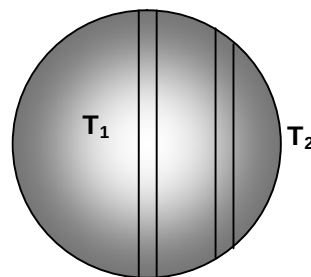
14. The given graph illustrates a transverse wave travelling on a string at a particular instant, and the points P, Q, R and S represent elements of the string. Which of the following statements about the motion of the elements is correct?



(A) The speed of the element at P is a maximum
(B) The displacement of the element at Q is always zero
(C) The energy of the element at R is entirely kinetic
(D) The magnitude of acceleration of the element at S is maximum.

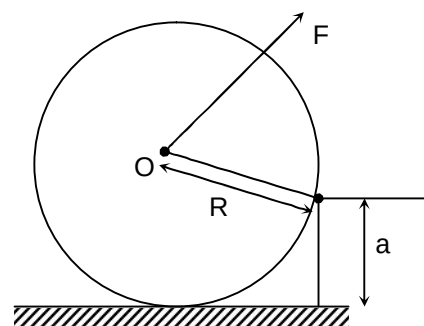
15. A tunnel is dug along the diameter of the earth. The time period of oscillation of a body dropped (from surface) in the tunnel is T_1 . If T_2 is the time period of similar body dropped (from surface) in a straight tunnel dug along any chord except through the centre of the earth, then

- (A) $T_2 < T_1$
 (B) $T_2 = T_1$
 (C) $T_2 > T_1$
 (D) any of A, B or C



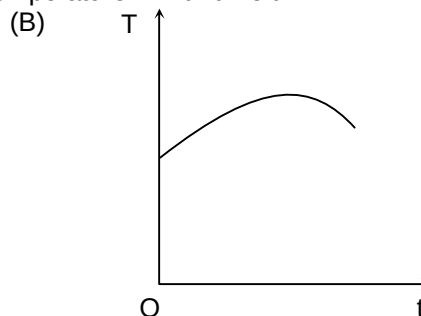
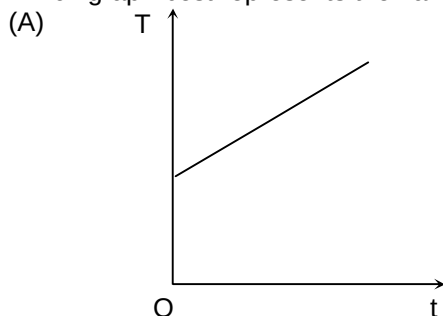
16. Suppose we consider a large number of containers each containing initially 10000 atoms of a radioactive substance with a half life of 1 year. After 1 year
- (A) all the containers will have 5000 atoms of the material,
 (B) all the containers will contain the same number of atoms of the material but that number will only be approximately 5000.
 (C) the containers will in general have different numbers of the atoms of the material but their average will be close to 5000.
 (D) none of the containers can have more than 5000 atoms.

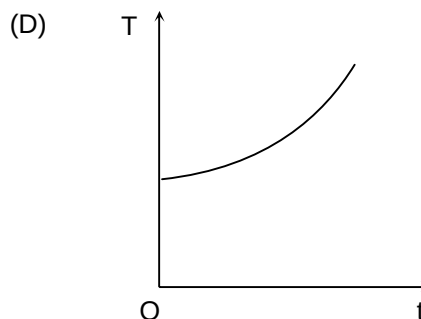
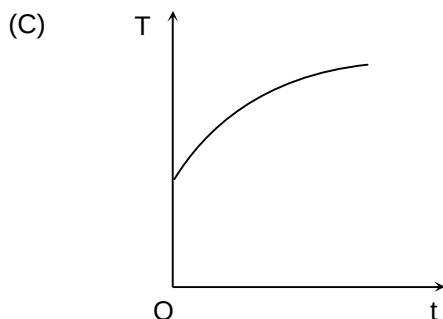
17. A uniform cylinder of mass M and radius R is to be pulled over a step of height a ($a < R$) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is:



- (A) $Mg\sqrt{1 - \frac{a^2}{R^2}}$
 (B) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2 - 1}$
 (C) $Mg\frac{a}{R}$
 (D) $Mg\sqrt{1 - \left(\frac{R-a}{R}\right)^2}$

18. The heat capacity C of a certain solid is related to its temperature T by the equation $C = aT^3$, where a is a constant characteristic of the solid. Heat is supplied to the solid at a steady rate. Which graph best represents the variation of its temperature T with time t ?



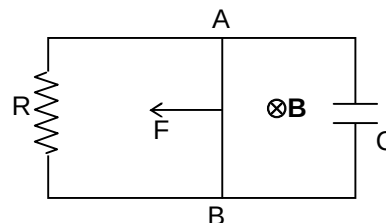


19. Two point masses, each of mass M are placed at $(L, 0)$ and $(-L, 0)$. A third equal point mass M , is uniformly rotating on the circle $x^2 + y^2 = L^2$. Equation of path traced by COM of system of these 3 point masses is
- (A) $x^2 + y^2 = L^2$
- (B) $x^2 + y^2 = \frac{L^2}{3}$
- (C) $x = y = 0$
- (D) $x^2 + y^2 = \frac{L^2}{9}$
20. If a proton is moving along the +ve x-axis at a speed v (small compared to speed of light), then find the magnetic field at $(a\hat{i} + b\hat{j})$, when the proton is at origin. Take charge on proton as q .
- (A) $\frac{\mu_0 qva}{4\pi(a^2 + b^2)} \hat{j}$
- (B) $\frac{\mu_0 qva}{4\pi(a^2 + b^2)^{3/2}} \hat{k}$
- (C) $-\frac{\mu_0 qva}{4\pi(a^2 + b^2)^{3/2}} \hat{k}$
- (D) $\frac{\mu_0 qva}{4\pi(a^2 + b^2)} \hat{k}$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A string of length 1m fixed at both ends is vibrating in 1st overtone. Tension in string is 200 N and linear mass density is 5 gm/m. Frequency (in Hz) of these vibrations is :
22. A conducting rod of mass m and of negligible resistance and length 0.5 m slides on a pair of parallel metallic rails, which are connected by a resistance $R = 10$ ohm and capacitance $C = 1\mu\text{F}$ as shown in the figure. A magnetic field $B = 1$ T is applied normal and into the plane of the circuit. At $t=0$, a force $F=0.4$ N is applied on the rod AB. The terminal speed (in m/s) acquired by the rod will be



23. When the voltage applied to an X-ray tube is increased from $V_1 = 15.5$ kV to $V_2 = 31$ kV, the wavelength interval between the K_α line and the cutoff wavelength of the continuous X-ray spectrum increases by a factor of $\frac{22}{19}$. If the atomic number of the element of the target is Z , the value of Z is (Take $hc = 1240$ eV-nm, Rydberg const = 10^7 m^{-1})
24. A cylindrical block of wood of base area 30 cm^2 , floats in a liquid of density 900 kg/m^3 . The block is depressed slightly and then released. The time period of the resulting oscillations of the block is equal to that of spring with block of same mass, then spring constant of this spring is equal to: (take $g = 10 \text{ m/s}^2$)
25. The current gain of a common emitter amplifier is 99. If the emitter current is 12 mA, then collector current in mA is
26. An unpolarized light of intensity 128 W/m^2 passes through three polarizers namely P_1 , P_2 and P_3 arranged one after the other such that P_1 is 1^{st} , P_2 is 2^{nd} and P_3 is last. The transmission axis of P_3 is parallel to that of P_1 . The intensity of light detected through the last polarizer is found to be 4 W/m^2 , then angle between the polarization axis of P_2 and P_3 in degrees is
27. A vernier caliper (Non standard) is constructed which has one main scale division of length 1 mm and 20 vernier scale divisions measure 18 mm. In an experiment, zero of vernier scale lies between 0.4 and 0.5 cm on main scale and 2^{nd} division of vernier scale coincides with a main scale division (when jaws of vernier are in closed position). When a cylinder whose diameter is to be measured is kept between the jaws of vernier caliper, the zero of vernier scale lies between 2.4 and 2.5 cm on main scale and 12^{th} division of vernier scale coincide with a main scale division. The diameter of cylinder in cm is
28. A parallel plate air capacitor has a capacitance C (figure-1). When it is half filled with a dielectric of dielectric constant 2 (figure-2), the percentage increase in the capacitance will be

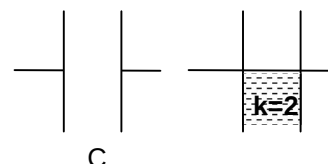
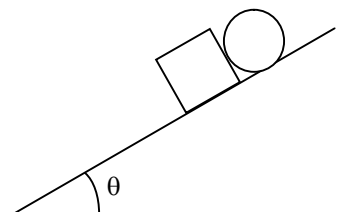


Figure-1

Figure-2

29. A block and a sphere of equal mass m are placed on an inclined plane. If the maximum frictional force that can exist between the block and the plane is equal to the weight of the block, and there is no frictional force between the sphere and the plane, what is the maximum angle θ at which the plane can be inclined before the block starts to slip (in degrees) is



30. The amplitude of a damped harmonic oscillator become half in 3 second and will become $1/x$ of the initial amplitude in next 6 second where x is :-

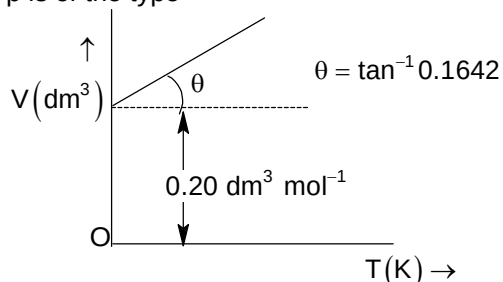
Chemistry

PART – B

SECTION – A (One Options Correct Type)

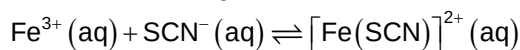
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. When intermolecular attractive forces are neglected, isobar formed between V and T at constant p is of the type



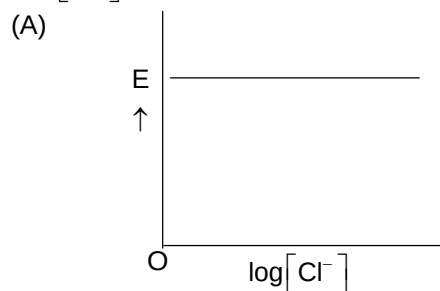
Pressure at which isobar is made,

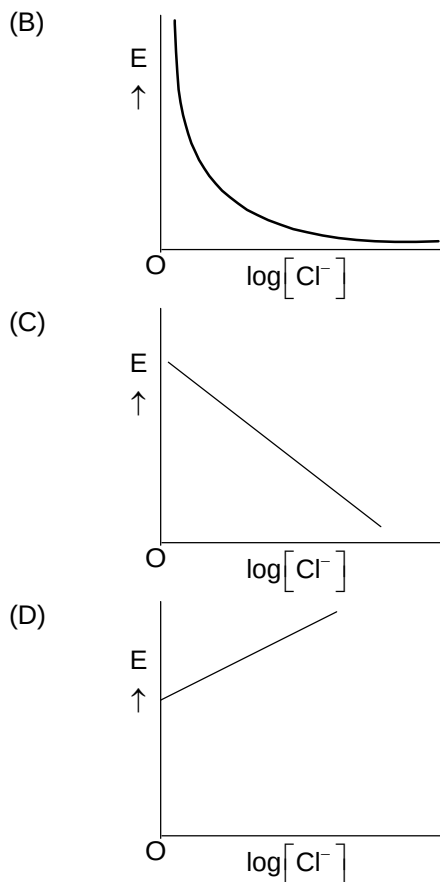
- (A) 2 atm
(B) 0.5 atm
(C) 1 atm
(D) 4 atm
32. Which of the following on the addition will not cause deep red colour to disappear?



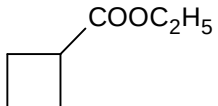
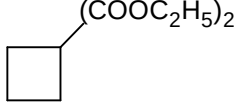
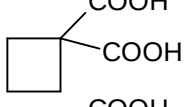
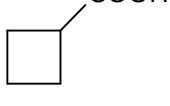
Pale yellow Colourless Deep red

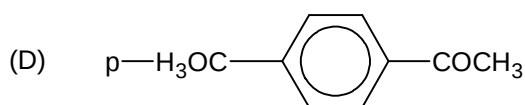
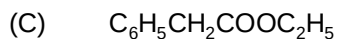
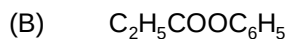
- (A) AgNO_3
(B) HgCl_2
(C) H_2O
(D) $\text{H}_2\text{C}_2\text{O}_4$
33. For the calomel half-cell, $\text{Hg}, \text{Hg}_2\text{Cl}_2 | \text{Cl}^{-}(\text{aq})$ values of electrode potentials are plotted at different $\log [\text{Cl}^{-}]$. Variation is represented by



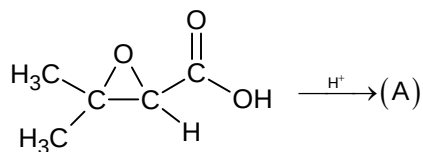


34. K_a of HCN and CH_3COOH are respectively 4.5×10^{-10} and 1.5×10^{-5} at 298 K. Thus, equilibrium constant for $\text{CN}^- + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{HCN}$ would be
- (A) 3.1×10^5
 (B) 3.2×10^{-5}
 (C) 3.3×10^{-4}
 (D) 3.3×10^4
35. Which of the following set of species have planar structures?
- (A) I_3^- , $\dot{\text{C}}\text{H}_3$, ClO_3^- , SiF_6^{2-}
 (B) I_3^+ , ICl_4^- , Al_2Cl_6 , TeCl_4
 (C) SCl_2 , N_2O_5 , SF_4 , XeOF_4
 (D) I_2Cl_6 , XeF_2 , BrF_4^- , XeF_5^-
36. $\text{XCl}_2 (\text{excess}) + \text{YCl}_2 \longrightarrow \text{XCl}_4 + \text{Y} \downarrow$;
 $\text{YO} \xrightarrow[>400^\circ]{\Delta} \frac{1}{2} \text{O}_2 + \text{Y}$; Ore of Y would be:
- (A) Siderite
 (B) Cinnabar
 (C) Malachite
 (D) Hornsilver

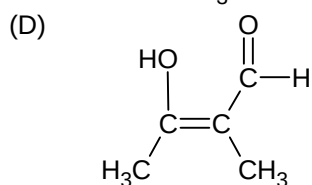
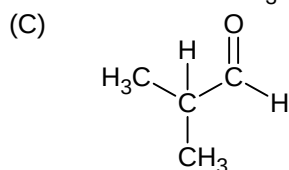
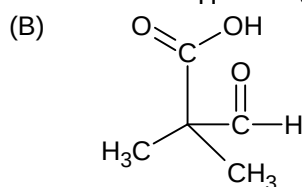
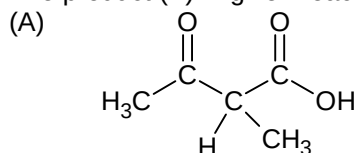
37. Nitrogen dioxide cannot be obtained from:
- $\text{Cu}(\text{NO}_3)_2$
 - $\text{Hg}(\text{NO}_3)_2$
 - NaNO_3
 - AgNO_3
38. Calcium imide on hydrolysis gives gas (B) which on oxidation by bleaching powder gives gas (C). Gas (C) on reaction with magnesium gives compound (D) which on hydrolysis gives again gas (B). Identify (B), (C) and (D)?
- $\text{NH}_3, \text{N}_2, \text{Mg}_3\text{N}_2$
 - $\text{N}_2, \text{NH}_3, \text{MgNH}$
 - $\text{N}_2, \text{N}_2\text{O}_5, \text{Mg}(\text{NO}_3)_2$
 - $\text{NH}_3, \text{NO}_2, \text{Mg}(\text{NO}_2)_2$
39. Select the correct statement about silicates:
- Cyclic silicate having three Si atoms contains six Si – O – Si linkages
 - $3\frac{1}{2}$ over oxygen atoms of per tetrahedron unit are shared in double chain silicate
 - $(\text{Si}_2\text{O}_5)^{2n-}_n$ is formula of double chain silicate
 - SiO_4^{4-} units polymerize to form silicate because Si atom has less tendency to form π – bond with oxygen
40. Which is not true regarding nylons?
- Usually a high melting solid polymer
 - Possesses a very high degree of crystallinity
 - Nylons are usually hydrophobic
 - Nylons have very high mechanical strength
41. What is the most likely product from the following reaction?
- $$\text{CH}_2(\text{COOC}_2\text{H}_5)_2 + \text{Br}(\text{CH}_2)_2\text{Br} \xrightarrow[\text{EtOH}]{\text{EtONa}} \xrightarrow[\Delta]{\text{H}^+/\text{H}_2\text{O}}$$
- 
 - 
 - 
 - 
42. An ester (A) with molecular formula $\text{C}_9\text{H}_{10}\text{O}_2$ was treated with excess of CH_3MgBr and the complex so formed, was treated with H_2SO_4 to give an olefin (B) whose ozonolysis gave a ketone with molecular formula $\text{C}_8\text{H}_8\text{O}$ which shows positive iodoform test. The structure of (A) is
- $\text{C}_6\text{H}_5\text{COOC}_2\text{H}_5$



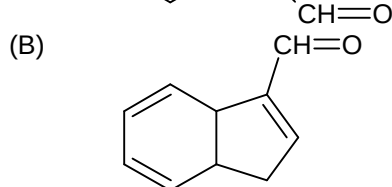
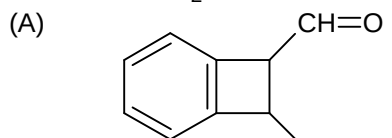
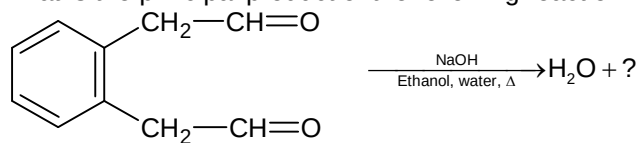
43.

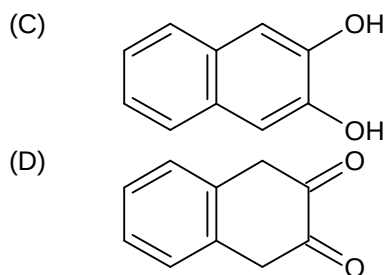


The product (A) in given reaction would be:

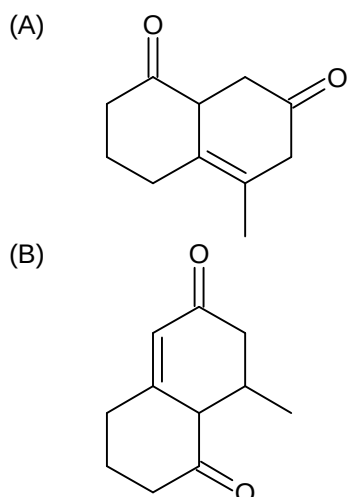
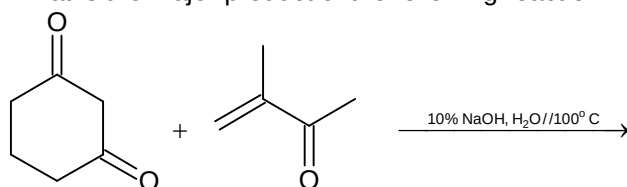


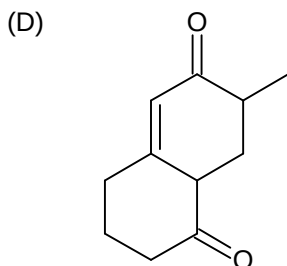
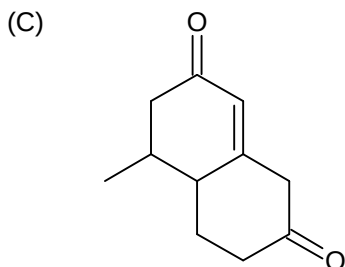
44. What is the principal product of the following reaction?





45. This structure of glycine (amino acid) is $\text{H}_3\text{N}^+\text{CH}_2\text{C}(=\text{O})\text{O}^-$ (Zwitter ion). Select the correct statement of the following?
- (A) Glycine, as well as other amino acids are amphoteric.
- (B) The acidic functional group in amino acids is $-\text{NH}_3^+$.
- (C) The basic functional group in amino acids is $-\text{CO}_2^-$.
- (D) All the statements are correct.
46. Select the incorrect statements about SO_2 :
- (A) It acts as oxidant and reductant both
- (B) It acts as bleaching agent but bleaching action is temporary
- (C) Its bleaching action is weak and that too by nascent H-atom
- (D) Its bleaching action is due to oxidation of coloured matter in presence of moisture
47. What is the major product of the following reaction?





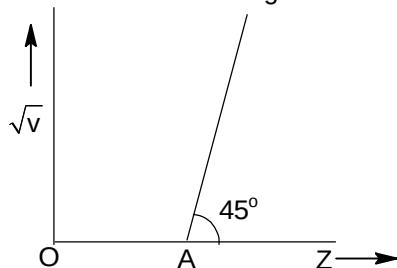
48. Hydrogen peroxide in its reaction with KIO_4 and NH_2OH respectively, is acting as a
- reducing agent, oxidizing agent
 - reducing agent, reducing agent
 - oxidizing agent, oxidizing agent
 - oxidizing agent, reducing agent
49. The reaction of white phosphorous with aqueous NaOH gives phosphine along with another phosphorous containing compound. The reaction type; the oxidation states of phosphorous in phosphine and the other product are respectively
- redox reaction; -3 and -5
 - redox reaction; $+3$ and $+5$
 - disproportionation reaction; -3 and $+1$
 - disproportionation reaction; -3 and $+3$
50. Select the incorrect statements
- Nitropentaammine chromium III chloride shows ionization isomerism
 - Among silver halides and calcium halides only AgF and CaF_2 are water soluble
 - In $\text{Fe}(\text{CO})_5$, the $\text{Fe}-\text{C}$ bond possesses both σ and π -character
 - Cu^+ and ethylene diamine gives a chelate compound due to multiple denticity

SECTION – B (Numerical Answer Type)

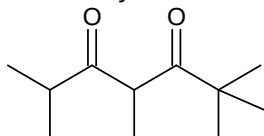
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The freezing point of an aqueous saturated solution of I_2 is -0.0024°C . More than this can dissolve in a KI solution because of the following equilibrium
- $$\text{I}_2(\text{aq}) + \text{I}^-(\text{aq}) \rightleftharpoons \text{I}_3^-(\text{aq})$$
- 0.1 M KI solution dissolves 12.5 g L^{-1} of I_2 . Calculate the equilibrium constant K_c for the above equilibrium. Assume molarity to be equal to molality and also assume that concentration of I in all saturated solutions is same. (K_f of water is $1.86 \text{ K molal}^{-1}$.)

52. Mosely equation is represented as $\sqrt{v} = a(Z-b)$, where a and b are constant. If $OA = 1$, calculate nuclear charge Z and 400 Hz. When frequency is 400 Hz



53. How many of the following species show paramagnetism?
 $\text{Na}^+, \text{O}_2, \text{Cu}^{2+}, \text{Fe}^{2+}, \text{Zn}^{2+}, \text{Cr}^{3+}, \text{Cu}^+, \text{O}_2^-, \text{O}_2^{2-}$
54. An ideal solution is made up of volatile liquids A and B for which $p_A^0 = 125$ Torr and $p_B^0 = 46.3$ Torr. As the pressure is reduced, first vapour is observed at 70.0 Torr. Calculate X_A .
55. Rate constant of decomposition of H_2O_2 to H_2O and O_2 is 0.0693 min^{-1} . Calculate volume of O_2 (in litres) obtained from the decomposition of 1.00 mole of H_2O_2 at 760 mm Hg and 293 K after 20.0 minutes.
56. Graph between $\log\left(\frac{x}{m}\right)$ and $\log p$ is a straight line inclined at an angle of 45° and $\log k = 0.6990$ for Freundlich's adsorption isotherm, $\left(\frac{x}{m}\right)$ is the amount of adsorbate adsorbed per gram of the adsorbent. What is the value of $\left(\frac{x}{m}\right)$ at 0.80 atm?
57. Consider the following complex compounds
 (I) $[\text{Pt}(\text{NH}_3)_2(\text{SCN})_2]$ (II) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
 (III) $[\text{Pt}(\text{en})\text{Cl}_2]$ (IV) $[\text{Cr}(\text{en})_2\text{Br}_2]^+$
 (V) $[\text{Rh}(\text{en})_3]^{3+}$ (VI) $[\text{CoCl}_2\text{Br}_2]^{2-}$
 Then calculate sum of total number of geometrical isomers in all above complex compounds.
58. 10 g of a mixture of hexane and ethanol are reacted with sodium to give 200 ml hydrogen at 27°C and 760 mm pressure. What is the percentage of ethanol into the mixture?
59. The mass adsorbed per g of adsorbed O_2 having pressure 10 atm at 400 K, if placed in contact with solid surface of 2 g in one litre vessel. The pressure of O_2 after adsorption becomes 2 atm. Assume no change in temperature and $R = 0.08 \text{ litre atm K}^{-1} \text{ mol}^{-1}$.
60. How many enol tautomers (including stereo isomers) can you draw for the following diketone?



Mathematics**PART – C****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If α is only real root of the equation $x^3 + \cos 1 \cdot x^2 + \sin 1 \cdot x + 1 = 0$, then $\tan^{-1} \alpha + \tan^{-1} \frac{1}{\alpha}$ is equal to
- (A) 0
(B) $\frac{\pi}{2}$
(C) $-\frac{\pi}{2}$
(D) π
62. $f(x)$ is a polynomial of degree 4, $f'(1) = f(2) = f(3) = f(4) = 0$, then root of $f(x) = 0$ is
- (A) $\frac{5}{11}$
(B) $-\frac{5}{11}$
(C) $\frac{11}{5}$
(D) $-\frac{11}{5}$
63. If a curve passing through (1, 1) satisfies the differential equation $x^2 dy = y^2 dx + xy d(xy)$, then the equation of the curve is
- (A) $\log|xy| = \frac{1}{x^2} - \frac{1}{y^2}$
(B) $\log|xy| = \frac{1}{x} - \frac{1}{y}$
(C) $-\log|xy| = \frac{1}{y^2} - \frac{1}{x^2}$
(D) $-\log|xy| = \frac{1}{x} - \frac{1}{y}$
64. The value of $S = \sum_{k=0}^n \frac{(-1)^k}{k^3 + 9k^2 + 26k + 24} \cdot {}^nC_k$ is
- (A) $\frac{1}{(n+3)(n+4)}$
(B) $\frac{1}{2(n+3)(n+4)}$
(C) $\frac{1}{2(n+1)(n+2)}$
(D) $\frac{1}{(n+1)(n+2)}$

65. Let complex number $z = (a + \cos \theta) + (2a - \sin \theta)i$. If $|z| \leq 2$ for any $\theta \in \mathbb{R}$, then the range of a is
- (A) $\left(-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$
- (B) $\left[-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right]$
- (C) $\left[-\frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right]$
- (D) $\left(-\frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$
66. The value of positive real parameter a such that the area of region bounded by parabolas $y = x - ax^2$, $ay = x^2$ attains its maximum value is equal to
- (A) $\frac{1}{2}$
- (B) 2
- (C) $\frac{1}{3}$
- (D) 1
67. Let S be the set of all those 2007-place decimal integers $2a_1a_2a_3 \dots a_{2006}$ which contain odd number of digit 9 in each sequence $a_1, a_2, a_3, \dots, a_{2006}$. The cardinal number of S is
- (A) $\frac{1}{2}(10^{2006} + 8^{2006})$
- (B) $\frac{1}{2}(10^{2006} - 8^{2006})$
- (C) $10^{2006} + 8^{2006}$
- (D) $10^{2006} - 8^{2006}$
68. Nine balls of the same size and color, numbered 1, 2, ..., 9, were put into a packet. Now A draws a ball from the packet, noted that it is of number a , and puts back it. Then B also draws a ball from the packet and noted that it is of number b . Then the probability for the inequality $a - 2b + 10 > 0$ to hold is
- (A) $\frac{52}{81}$
- (B) $\frac{59}{81}$
- (C) $\frac{60}{81}$
- (D) $\frac{61}{81}$
69. If $L = \lim_{x \rightarrow 4} \left(\frac{\log_5 \left(\frac{5x}{4} \right)}{\log_5 \frac{x}{4}} \right)^{\left(\log_5 \frac{x}{4} \right)^{-1}}$, then $[L]$ is (where $[.]$ denotes greatest integer function)
- (A) 3
- (B) 7
- (C) 9
- (D) 12

70. A tower subtends angles α , 2α and 3α respectively at points A, B and C, all lying on a horizontal line through the foot of the tower. Then AB/BC is equal to
- (A) $\frac{\sin 3\alpha}{\sin 2\alpha}$
 (B) $1 + 2 \cos 2\alpha$
 (C) $2 \cos 2\alpha$
 (D) $\frac{\sin 2\alpha}{\sin \alpha}$
71. The SD of 15 items is 6 and if each item is decreases by 1, then standard deviation will be
- (A) 5
 (B) 7
 (C) $\frac{91}{15}$
 (D) 6
72. $\sim(p \vee q) \vee (\sim p \wedge q)$ is logical equivalent to
- (A) $\sim p$
 (B) p
 (C) q
 (D) $\sim q$
73. If matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, then which of the following is in correct?
- (A) $A^3 - A^2 = A - I$
 (B) $|A^{2022} - I| = 0$
 (C) $A^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$
 (D) $A^{50} = \begin{bmatrix} 1 & 1 & 0 \\ 50 & 1 & 0 \\ 50 & 0 & 1 \end{bmatrix}$
74. The distance of the point P(-1, -2, 4) from the plane $x + y + z = 7$ measured parallel to z-axis is
- (A) 4
 (B) 3
 (C) 5
 (D) 6
75. The sum of all possible real values of 'a' for which vectors $\vec{r}_1 = \ln|a|\hat{i} + a\hat{j} - \hat{k}$ and $\vec{r}_2 = (1+a^2)\hat{i} - \hat{j} - (a-1)\hat{k}$ are orthogonal is
- (A) -2
 (B) 0
 (C) 2
 (D) 4

76. Let f be a continuous function that $p(x) = \int_{-2}^x f(t)|x-t|dt$ where $x \in (-2, 2)$, then the value of $\int_0^2 \frac{p''(x)}{f(x)} dx$ is equal to
- (A) 0
(B) 1
(C) 2
(D) 4
77. A line L_1 passing through point $A(2, 0)$ and makes an angle θ with positive x -axis in anti-clockwise direction, where $\tan \theta = \frac{1}{2}$. Now, L_1 is rotated about the point A in anti-clockwise direction through an angle of $(\pi - 4\theta)$. If the line in the now position is L_2 . Then the positive slope of the angle bisector between the lines L_1 and L_2 is
- (A) 2
(B) $\frac{3}{2}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$
78. If $\tan(\alpha - \beta) = \frac{\sin 2\beta}{3 - \cos 2\beta}$, then $\tan \alpha = f(\beta)$. The value of $f\left(\frac{\pi}{4}\right)$ equals
- (A) 2
(B) $\frac{1}{2}$
(C) 3
(D) $\frac{1}{3}$
79. In a parabola $y^2 = 4ax$, two points P and Q are taken such that the tangents drawn to parabola at these points meet at directrix in R . Focus of locus of circumcentre of $\triangle PQR$ will be
- (A) $\left(\frac{a}{2}, 0\right)$
(B) $(a, 0)$
(C) $\left(\frac{3a}{2}, 0\right)$
(D) $\left(\frac{5a}{2}, 0\right)$
80. If $f(x) = \int \frac{\tan^3 x}{2 + \tan^2 x} dx$, then
- (A) $f(x) = \log \left| \frac{2 + \sin^2 x}{\cos x} \right| + C$
(B) $f(x) = \log \left| \frac{2 - \sin^2 x}{\cos x} \right| + C$

$$(C) \quad f(x) = \log \left| \frac{2 - \cos^2 x}{\sin x} \right| + C$$

$$(D) \quad f(x) = \log \left| \frac{2 + \cos^2 x}{\sin x} \right| + C$$

SECTION – B
(Numerical Answer Type)

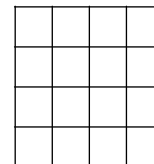
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81. The number of integral points (i.e., the points whose x and y-coordinates are both integers) within the area (not including the boundary) enclosed by the right branch of hyperbola $x^2 - y^2 = 1$ and the line $x = 100$ is

82. Let $f(x)$ be an odd function on $\mathbb{R}$, and $f(x) = x^2$ for $x \geq 0$. Suppose for any $x \in [a, a + 2]$, $f(x + a) \geq 2f(x)$. Then the range of real number a is $[\sqrt{\alpha}, \infty)$. Then α is

83. The largest possible value of the expression $S = \sin x_1 \cdot \cos x_2 + \sin x_2 \cdot \cos x_3 + \dots + \sin x_{99} \cdot \cos x_{100} + \sin x_{100} \cdot \cos x_1$, where $x_1, x_2, x_3, \dots, x_{100}$ are arbitrary real numbers is

84. Four letters, two 'a's and two 'b's, are filled into 16 cells of a matrix as shown in figure. It is required that each cell contains at most one letter, and each row or column cannot contain the same letters. Then there are _____ different ways that the matrix can be filled



85. Given that $|z| = 1$ and satisfies $\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$. Then the number of solution of the equation is

86. Suppose a and b are positive real numbers satisfying $\frac{1}{a} + \frac{1}{b} \leq 2\sqrt{2}$ and $(a - b)^2 = 4(ab)^3$. Then $|\log_a b|$ is equal to

87. Let a, b be real numbers, and $f(x) = ax + b$ satisfies $|f(x)| \leq 1$ for any $x \in [0, 1]$. Then the maximum of ab is

88. Let $n = \text{aaaaaa}$ be a six digit number (all digit repeated) and N is divisible by 924. Let a and b are the roots of the equation $x^2 - 11x + \lambda = 0$, then the sum of possible values of λ is

89. Let ABC be a triangle with angle $B = \frac{\pi}{4}$ and $c = 8$. A circle with centre A and radius AB meets BC at D . If altitude from vertex A to the side BC meets the circle at E . Then area of the $\triangle BED$ is $\alpha(\sqrt{\beta} - 1)$. Then the value of $\frac{\alpha}{\beta}$ is

90. Given 2022 vectors on a plane. Sum of every 2021 vectors is a scalar multiple of the other vector. Not all vectors are scalar multiple of each other. The magnitude of sum of all these vectors is